

Homework 4

Instructions

Complete the following problems and submit to Gradescope your work as a file named `hw4.Rmd`.

A file with the wrong name or a bug will prevent the autograder from being able to run your submission. Check that your filename is correct, and check for bugs clearing your memory and then running your file from start to finish. It's also a good idea to check that your file "knits."

Note: Do not use any loops and hard-code as little as possible.

Problem 1

The data file `fatal_accidents.csv` contains a record of each fatal accident in Virginia and surrounding states in 2021. Each record details the number of vehicles involved, the number of people involved in each accident - both inside and outside the vehicle, and timing information for each fatal accident. The day of the week begins with 1 for Sunday.

a

Read in the file and save the data as a data frame called `fatal`.

```
str(fatal)
```

```
## 'data.frame':  3260 obs. of  11 variables:
## $ State      : chr  "District of Columbia" "District of Columbia" "District of Columbia" "Dist.
## $ Case.number : int  110001 110002 110003 110004 110005 110006 110007 110008 110009 110010 ...
## $ Vehicle.count : int  1 2 1 1 2 1 2 1 1 1 ...
## $ People.count.IN : int  1 4 1 1 2 1 3 1 3 1 ...
## $ People.count.OUT: int  1 0 1 1 0 1 0 1 1 1 ...
## $ Day         : int  12 21 7 3 23 1 16 1 1 3 ...
## $ Month        : int  1 1 2 2 2 3 3 3 4 4 ...
## $ Year         : int  2021 2021 2021 2021 2021 2021 2021 2021 2021 2021 ...
## $ Day.of.week  : int  3 5 1 4 3 2 3 2 5 7 ...
## $ Hour         : int  23 23 20 20 18 11 23 15 15 0 ...
## $ Minute       : int  22 52 2 2 9 7 10 18 15 34 ...
```

b

Create a list called `state.list` that separates the data set by state. Each element of the list should contain a data frame subset to a single state. Each element should be named with the name of the state.

Do NOT print the list.

```
head(state.list[['Virginia']])
```

```
##           State Case.number Vehicle.count People.count.IN People.count.OUT Day
## 2098 Virginia      510001           1           3           0  1
## 2099 Virginia      510002           1           2           0  2
## 2100 Virginia      510003           1           1           1  6
## 2101 Virginia      510004           1           1           0  6
## 2102 Virginia      510005           1           2           0  6
## 2103 Virginia      510006           1           2           0  5
##           Month Year Day.of.week Hour Minute
## 2098          1 2021           6     1     50
## 2099          1 2021           7    22     22
## 2100          1 2021           4     6     44
## 2101          1 2021           4    10     47
## 2102          1 2021           4    15     50
## 2103          1 2021           3    13      5
```

c

Use an appropriate `apply()` function to obtain the first three rows of each element of the list created in part b. Call this list `listBabyChunks`

```
listBabyChunks[[1]]
```

```
##           State Case.number Vehicle.count People.count.IN
## 1 District of Columbia      110001           1           1
## 2 District of Columbia      110002           2           4
## 3 District of Columbia      110003           1           1
## People.count.OUT Day Month Year Day.of.week Hour Minute
## 1           1 12     1 2021           3    23     22
## 2           0 21     1 2021           5    23     52
## 3           1  7     2 2021           1    20      2
```

d

Use an appropriate `apply()` function with `state.list` to create a `list` that contains tables that show the percentage (not proportion!) of accidents in 2021 that occurred on each day of the week. Each element of the list should contain the `table` for a single state. Call this variable `accidentsBreakdown`

Note: The percentage values should be rounded to one decimal. For example, 95.1% will be displayed as 95.1.

```
accidentsBreakdown[['West Virginia']]
```

```
##
##      1      2      3      4      5      6      7
## 12.5 11.7 12.1 12.8 16.3 21.0 13.6
```

e

What observations can you draw from the tables created in part d? Write your answer in markdown with a clear heading called “Question 1 Part E Free Response”.

f

Using an appropriate `apply()` function with `state.list`, create a `list` that contains tables that show the number of accidents in 2021 that occurred on each day of the week (rows) and the number of vehicles involved (columns). Each element of the list should contain the table for a single state. Call this variable `accidentsBreakdown2`

```
accidentsBreakdown2[['District of Columbia']]
```

```
##           Vehicle.count
## Day.of.week 1 2 3 4 5
##           1 6 1 0 0 0
##           2 5 1 1 1 0
##           3 2 2 0 0 0
##           4 2 0 0 0 0
##           5 1 2 2 0 0
##           6 1 3 0 1 0
##           7 6 2 0 0 1
```

g

What changes to the presentation of the tables in part f might improve ease of drawing observations from them? Write your answer in markdown with a clear heading called “Question 1 Part g Free Response”.

Problem 2

Continue using the data file `fatal_accidents.csv` described in problem 1. For the following, use functions in the `tidyverse`

a

Create a copy of the data frame `fatal`. Call it `fatal2`. Add a column to this data frame called `People.count` that contains the total number of people involved in each accident.

```
head(fatal2)
```

```
##           State Case.number Vehicle.count People.count.IN
## 1 District of Columbia    110001           1           1
## 2 District of Columbia    110002           2           4
## 3 District of Columbia    110003           1           1
## 4 District of Columbia    110004           1           1
## 5 District of Columbia    110005           2           2
## 6 District of Columbia    110006           1           1
##  People.count.OUT Day Month Year Day.of.week Hour Minute People.count
## 1              1  12      1 2021           3    23     22           2
## 2              0  21      1 2021           5    23     52           4
## 3              1   7      2 2021           1    20      2           2
## 4              1   3      2 2021           4    20      2           2
## 5              0  23      2 2021           3    18      9           2
## 6              1   1      3 2021           2    11      7           2
```

b

Calculate the average number of vehicles and the average number of people (columns) involved in accidents in 2021 in each state (rows) in a single two-dimensional data structure. The ordered column names should be "State", "Vehicle.mean" and "Person.mean". The overall variable should be called `byState`

```
head(byState)
```

```
## # A tibble: 5 x 3
##   State          Vehicle.mean Person.mean
##   <chr>          <dbl>         <dbl>
## 1 District of Columbia      1.68         2.75
## 2 Maryland                  1.60         2.42
## 3 North Carolina            1.58         2.37
## 4 Virginia                  1.55         2.18
## 5 West Virginia             1.51         2.15
```

c

Calculate the minimum, average, and maximum number of vehicles involved in accidents in 2021 in each state (rows) in a single two-dimensional data structure. Call your variable `byState2`. Its column names should be "State", "Vehicle.min", "Vehicle.mean" and "Vehicle.max".

```
head(byState2)
```

```
## # A tibble: 5 x 4
##   State          Vehicle.min Vehicle.mean Vehicle.max
##   <chr>          <int>         <dbl>         <int>
## 1 District of Columbia      1         1.68           5
## 2 Maryland                  1         1.60           6
## 3 North Carolina            1         1.58          10
## 4 Virginia                  1         1.55           6
## 5 West Virginia             1         1.51           5
```

d

What observations can you draw from the tables created in parts b and c? Write your answer in markdown with a clear heading called "Question 2 Part D Free Response".

e

Display the number of accidents (columns) that occurred in Virginia in each month (rows) of 2021 in a single two-dimensional data structure. Call your variable `virginiaByMonth`

It should look like this:

```
virginiaByMonth
```

```
## # A tibble: 12 x 2
##   Month Month.total
##   <int>         <int>
```

```
## 1      1      50
## 2      2      47
## 3      3      57
## 4      4      83
## 5      5      87
## 6      6      65
## 7      7     100
## 8      8      77
## 9      9      79
## 10     10     108
## 11     11      79
## 12     12      74
```

f

Calculate the median and mean number of vehicles involved in accidents that occurred in Virginia on each day of the week in each of the summer months (June through August) of 2021 in a single two-dimensional data structure. Call your variable `byMonthAndWeekday`

It should look like this

```
byMonthAndWeekday
```

```
## # A tibble: 21 x 4
## # Groups:   Month [3]
##   Month Day.of.week Vehicle.med Vehicle.mean
##   <int>      <int>      <dbl>      <dbl>
## 1     6          1          1          1.43
## 2     6          2          2          1.71
## 3     6          3          2          1.78
## 4     6          4          1          1.3
## 5     6          5          1          1.42
## 6     6          6          2          1.73
## 7     6          7          2          1.6
## 8     7          1          1          1.44
## 9     7          2          1          1.5
## 10    7          3          1.5          1.9
## # i 11 more rows
```

g

What observations can you draw from the tables created in parts e and f? Write your answer in markdown with a clear heading called “Question 2 Part G Free Response”.